UNIVERSITY OF BIRMINGHAM | GRAVITATIONAL WAVE ASTRONOMY

OBSERVATIONAL COSMOLOGY

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RECAP: COSMOLOGICAL CONSTANT

- The Friedmann equations lead to an expanding or contracting universe
- To obtain a static solution, Einstein introduced the **cosmological constant** Λ (unstable!)

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = H(t)^2 = \frac{8\pi G\rho(t)}{3} - \frac{\kappa c^2}{R_0^2 a^2(t)} + \frac{\Lambda}{3}$$

- Adding this term is equivalent to adding another "fluid" component <u>with a constant energy density</u>

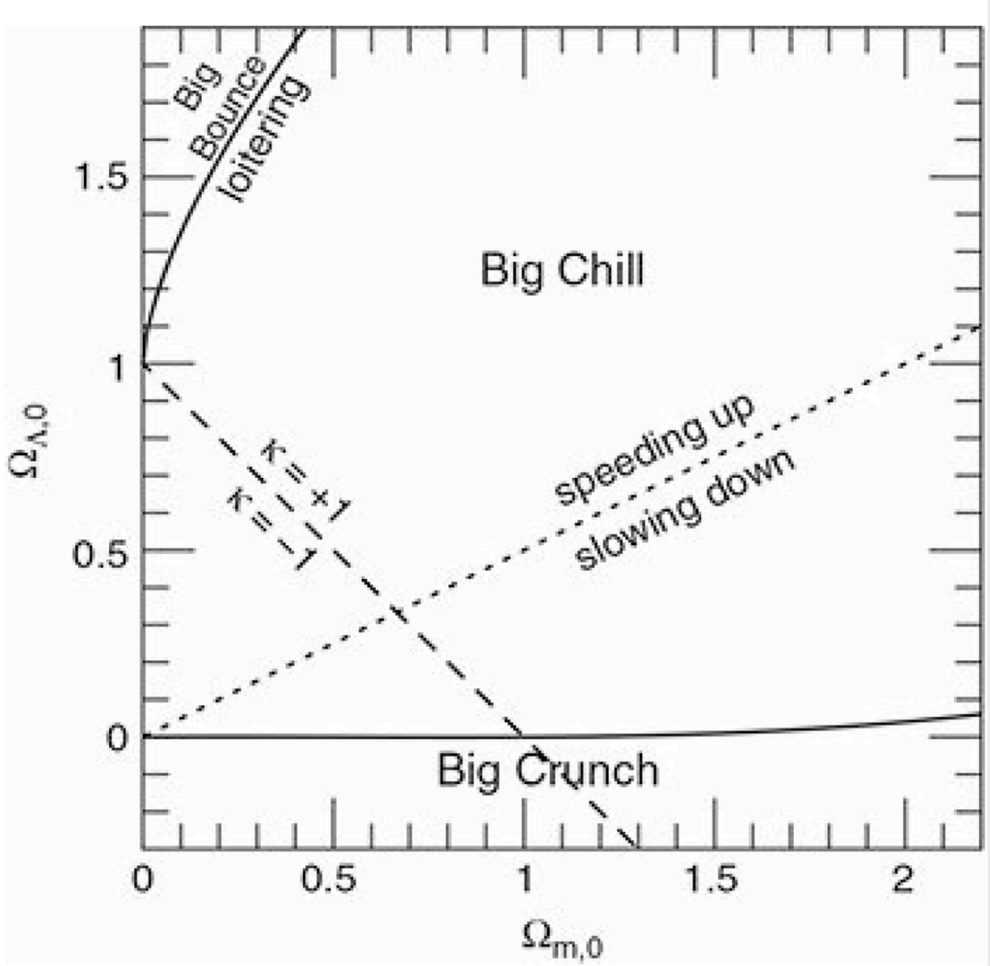
$$\rho_\Lambda = \frac{\Lambda}{8\pi G} = \text{const.}$$

- From the fluid equation we find

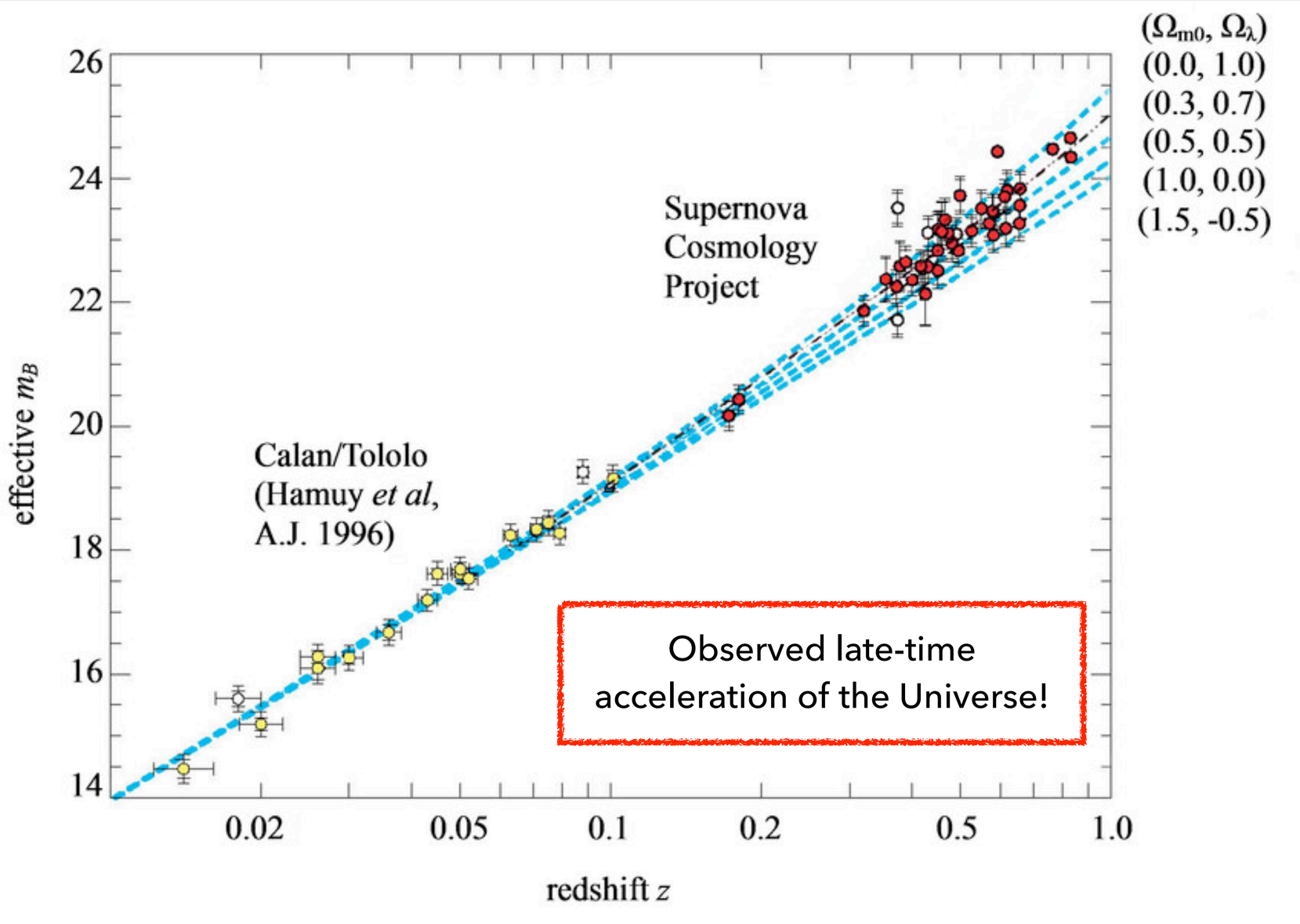
$$p_\Lambda = -\rho_\Lambda c^2 \quad \Rightarrow w = -1$$

- "fluid" with <u>negative pressure</u> that counteracts the deceleration of the expansion

Big effect on possible evolutionary paths!



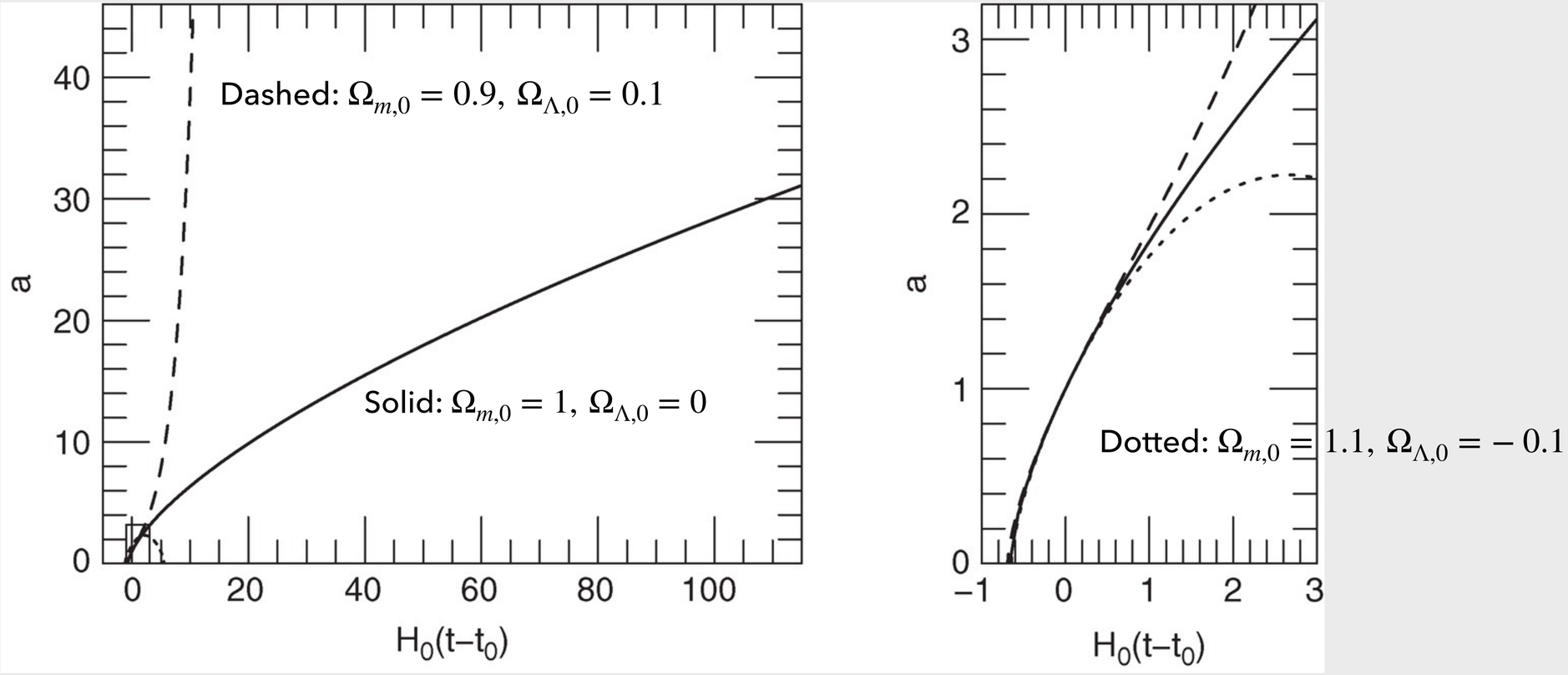
WHY Λ?


$(\Omega_{m0}, \Omega_\lambda)$
(0.0, 1.0)
(0.3, 0.7)
(0.5, 0.5)
(1.0, 0.0)
(1.5, -0.5)
Supernova Cosmology Project
Calan/Tololo (Hamuy *et al*, A.J. 1996)
Observed late-time acceleration of the Universe!
effective m_B
redshift z
26
24
22
20
18
16
14
0.02
0.05
0.1
0.2
0.5
1.0

FLAT Λ+MATTER MODEL UNIVERSE

- Age of such a universe:

$$t_0 = \frac{2}{3\sqrt{1-\Omega_{m,0}}H_0} \ln\left(\frac{1+\sqrt{1-\Omega_{m,0}}}{\sqrt{\Omega_{m,0}}}\right)$$



[Fig. 5.5 from Ryden]